Slayer 1.07: data and training method

What the model should be able to do, what data and why, how we train

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Abstract. We present a dataset and a plan for the further pretraining of the Slayer model with about 1,07 billion parameters. The starting point, Slayer 1.06, consumed 14 520 680 448 tokens in 13 848 steps. The prepared 1.07 set contains 8 891 005 707 tokens across 9 files, encoded with Slayer's tokenizer; it covers Polish web, English educational and mathematical texts, as well as a replay of the Polish core and legal data. We describe personal-data masking, the removal of documents containing fragments of tests, and verification of token consistency with the source. After review we separate the improvement of the final model from the causal effect of the new data: control covers only the common 150-step segment and the available results of the reference arm. The protocol separates checkpoint selection from evaluation, treats letter accuracy as a diagnostic, and accounts for checkpoint noise and multiple comparisons. In the completed short probe, a format insert did not remove the letter's dominance. The domain insert met the criterion of a downward movement of the BPB of the correct answer, but such movement also covered the control. The predicted domain specificity was not confirmed; we do not equate this with an absence of insert effect. The 1.07 stage training was not carried out. The selection of further sources, the release configuration and the launch remain decisions of the training team. This document does not present results of the 1.07 model.

1 Goal: what Slayer should be able to do

Project assumptions.

- **Volume:** a minimum of 40 billion training tokens for the 1B model, targeting 100 billion. The prepared 1.07 stage covers 8 891 005 707 tokens, including 1 894 295 941 tokens of targeted replay of the Polish 1.06 data. We do not equate the number of training exposures with the volume of unique texts. The timing and configuration of the run are chosen by the training team after reviewing the data and this document.
- **Languages:** Slayer started from Polish. Further data may have more English, but not by force: the EN share follows from the fact that there is much less Polish text than English.
- **Skills:** the model should learn skills, also by methods that worked elsewhere. We adopt a method only when it has evidence in the literature or in our measurement (section 6).

Starting point: Slayer 1.06. The realized composition of the 1.06 data files according to the manifests: web 53,9 %, core 18,6 %, legal 9,2 %, EN 18,2 %. These are not the target sampling weights 63/19/8/10 from the dataset card. The sum of the file contents is 14 524 133 431 tokens, whereas the number of tokens consumed in training is $13\,848 \times 1\,048\,576 = 14\,520\,680\,448$. This gives about 13,5 tokens per parameter. The value of about 20 tokens per parameter is a reference point for the compute budget, and not a guaranteed threshold for attaining a skill. Observations for 1.06 are described in section 2. In brief:

1. in the studied multiple-choice question protocols Slayer selects a single letter in 83–94 % of questions; letter accuracy alone does not separate format preference from skill;
2. to short-answer questions it responds with a full sentence, so exact match yields almost 0 (0,005 on NQ and 0,015 on TriviaQA on the panel of 200);
3. after calibrating the letter preference it stays close to the random level on MMLU and MedMCQA; at least two hypotheses are consistent with this: insufficient domain data or undertraining, which this result does not separate;
4. there is little Polish domain (textbook) knowledge in the data.

What 1.07 should improve, in order of importance.

1. **Knowledge (PL and EN):** check whether further training on educational and domain texts improves tasks outside training (section 3). The verdict family and the metrics are defined by section 9; letter accuracy and the BPB of the correct answer remain diagnostics, not standalone evidence of a knowledge gain.
2. **Maintaining Polish:** with a larger EN share Polish must not deteriorate beyond noise. Measures: heldout PL, MultiBLiMP-PL, ARC-Easy-PL. A drop beyond noise is a cost that we report explicitly (section 9).
3. **Answer format:** measure the A–D letter and the short answer separately. The F/K data are not part of the prepared 1.07 set. The completed probe separates the success of the BPB movement from confirming domain specificity; its results and limitations are described in section 7. The decision on add-ons remains with the training team.

What this document does not promise. The magnitudes of the gains are hypotheses recorded before training, not results. Tasks on which both Slayer and LFM2.5 guess on our 200-item panel have low priority at the current budget. This does not, however, prove that they

cannot be improved with data. Tasks with leakage of test sources into the training data do not enter the conclusions (section 4).

2 Starting state: measured shortcomings of Slayer 1.06

The reference point is Slayer 1.06 at step 13 848. The comparison model of the same scale is LFM2.5-1.2B-Base. The numbers come from the bench50 panel (200 items per task, `results.json` Space `1cf2b8bb`) and from the saved per-item results. We consider a result distinguishable from guessing when the lower bound of CI95 exceeds max(chance level, majority-class share). For tasks with subsets (MMLU, BLiMP, BBH) we compute the CI with a Rao–Wu bootstrap.

Table 1: bench50 tasks by distinguishability from guessing (panel $n = 200$).

class	tasks	examples
Slayer at random level, LFM2.5 above	8	ARC-C, WinoGrande, SST-2, PolEmo2-in, MMLU, RACE, MedMCQA, PAWS
both models at random level	13	BoolQ, CB, WiC, RTE, ANLI-r3, PROST, Belebele-PL, PPC, PSC, CBD, KLEJ-NER, LLMzSzŁ, CoLA
both above random level	8	BLiMP, MultiBLiMP-PL, HellaSwag, OpenBookQA, SciQ, COPA, 8tags, DYK
without random baseline or without result	21	generation, perplexity

1. Letter format. In A–D questions Slayer 1.06 selects the dominant letter in 83–94 % of answers, and Pythia-1B in 84,4–96,0 %. The letter accuracies of both models differ only by noise (benchmarks-fabryka-ai repository, `research/analyses/answer-format`). On LLMzSzŁ Slayer selects B in 66 % of questions (internal measurement). We therefore treat the MMLU, MedMCQA and Belebele-PL results computed from letters as a format artifact, and not as a measure of knowledge. By answer content (cloze, ARC-C acc_norm, paired results) the entire gain between versions falls on the 1.00→1.03 segment: +3,84 pts, $z = 2{,}19$. The 1.03→1.06 segment, i.e. about 7 billion tokens, gives +0,17 pts, $z = 0{,}09$, i.e. noise (the same source). The 1.06 result, 0,2577, lies below the majority baseline 0,265.

2. Knowledge after subtracting format. After subtracting the letter preference (leave-one-out calibration, without labels; internal measurement) the dominant-letter share drops to 27–36 %, but the accuracy on MMLU, MedMCQA, LLMzSzŁ and Belebele-PL stays at the random level. The format alone therefore does not reveal hidden knowledge. We have two explanations and 1.07 does not separate them: *lack of domain knowledge in the data* or *undertraining* (1.06 is about 14,5 billion tokens without LR decay). Pythia-1B after 300 billion tokens, i.e. about 20 times more, has MMLU from letters 0,231 and ARC-C from letters 0,229, so also the random level (`results/LICZBY.json` of the same repository). The random level of a 1B model on these tasks is therefore typical at a budget on the order of hundreds of billions of tokens and by itself does not indicate what is missing.

3. Short answer. On NQ-Open and TriviaQA Slayer has exact_match 0,005 and 0,015 (LFM2.5: 0,080 and 0,310). It responds with a full sentence, and some of the hits in the "contains the answer" scoring are an echo of the question. After subtracting it the margin is about +5 pts on TriviaQA (estimate).

4. Constant answer in classification. Calibration of the label preference shows that BoolQ, CB, WiC, PSC, CBD and CoLA carry no signal either in Slayer or in LFM2.5. The exception is sentiment: SST-2 after calibration rises from 0,505 to 0,585 in Slayer (a weak signal, $z \approx 2{,}3$) and to 0,855 in LFM2.5.

5. Grammar. BLiMP: 62 of 67 paradigms above 0,5. Four of the remaining five are below or at the random level also in Pythia-1B, so we treat them as a feature of scale. The fifth (`matrix_question_npi_licensor_present`: Slayer 0,497, Pythia 0,632) is a gap for Slayer, but part of the errors there follows from the instrument: the sum of logprob rewards the shorter sentence. MultiBLiMP-PL: 0,99.

What this implies for 1.07. The tasks from the first row of tab. 1 are data targets, because they are solved by a model of this scale at a much larger token budget. LFM2.5-1.2B-Base was trained on 28 trillion tokens (model card on HF), i.e. about 1 900 times more than 1.06. The difference against LFM2.5 therefore measures budget and data together, and not the model scale alone. The tasks from the second row have low priority at the current budget. The current panel did not show a skill there in any model, but this does not prove that they cannot be improved with data. Numbers on tasks with leakage hits in the 1.06 corpus (marked on the Space as *flagged*) we read only on positions without hits (section 4).

2.1 1.06 numbers: bench50 panel and track

bench50 panel (200 items per task, protocol `bench50-v0.3.1-panel200`). Selected tasks concerned by the shortcomings from this section. Reference: LFM2.5-1.2B-Base in the same protocol. The "without hits" column is Slayer's result on panel positions without a 13-gram in the 1.06 corpus (scan of the 1.06 corpus).

task	metric	Slayer 1.06	without hits (n)	LFM2.5	random level
T02 ARC-C	acc_norm	0,250	0,250 (196)	0,490	0,25
T32 MMLU	acc	0,283	0,294 (194)	0,563	0,25
T45 MedMCQA	acc	0,245	0,241 (191)	0,405	0,25
T37 LLMzSzŁ	acc	0,300	0,298 (191)	0,290	0,25
T24 Belebele-PL	acc	0,300	—	0,300	0,25
T05 HellaSwag	acc_norm	0,365	0,365 (200)	0,625	0,25
T42 NQ-Open	exact_match	0,005	0,005 (195)	0,080	—
T43 TriviaQA	exact_match	0,015	0,016 (188)	0,310	—
T01 BLiMP	acc	0,841	0,838 (197)	0,801	0,50

At $n = 200$ and $p \approx 0{,}3$ the standard error of a single result is about 0,03. The CI95 intervals published on the results page are too narrow: the stratified bootstrap underestimated the variance by a factor of $(n_h - 1)/n_h$. So far we have recomputed three tasks (T01, T32 and T47; the first two are in the table): their corrected intervals are 1,09–1,32 times wider. The remaining tasks will be recomputed only by the page fix. The verdicts and the Holm correction do not change, because p comes from a permutation test. The fix has not yet landed, which is why here we give only the values, without intervals.

Track (protocol v8), 1.06 = step 13 848. ARC-Easy 52,99 %, BLiMP 76,12 %, WikiText-2 byte-PPL 2,2849, MultiBLiMP-pl 98,41 %, ARC-Easy-PL EU20 45,03 %. These are numbers from a different protocol than the bench50 panel (e.g. BLiMP: here the full 67 000 pairs, there a panel of 200 items with different code), so we do not combine the two tables (section 9).

3 Data: composition of ca. 9 billion tokens and rationale

3.1 Starting point: what 1.06 saw

Slayer 1.06 finished at step 13 848, so training consumed $13\,848 \times 1\,048\,576 = 14\,520\,680\,448$ tokens (13.52 tokens per parameter at $P = 1\,073\,912\,851$). The eight input-set files contain in total 14 524 133 431 training tokens: PL web 7.835 bn, PL core 2.707 bn, PL legal texts 1.339 bn, EN 2.643 bn. EN is therefore 18.2 % of v6 tokens. The dataset card on HF reports 63/19/8/10 (EN 10 %), but this is the *target weights* of the input-set builder (the `wagi` field in the manifests), not the realized share in v6 tokens (the `tokeny_v6_na_komponent` field). In this text we use realized shares. Every core, legal, and EN document was seen exactly once (the intervals in the manifests sum to one epoch, with no gaps and no overlap), and the core, legal, and EN pools are exhausted: there is no fresh text of these kinds in our corpus.

3.2 Composition of 1.07

The goal of this stage is ca. 9 billion new tokens, as the first step of the path to at least 40 bn in total. We selected the composition according to three principles:

1. **Polish stays a constant component** (replay). The EN share in the new data rises to 44 % (from 18 % in 1.06), so we replay part of the core and legal data. The paper on continued training in another language (arXiv:2403.08763) studies a full language switch EN→DE, whereas for us Polish is still ca. 70 % of the whole diet. Replay is therefore a precaution here, not a requirement following from that paper.
2. **We exclude exact EN replays.** An EN document whose text key (`sha256` of the normalized text, 16 hex characters) is in the EN corpus of 1.06 is dropped. The same applies to duplicates within the build.
3. **The text undergoes the same processing as in 1.06:** personal-data masking with the `maskuj()` function from the 1.06 builder (import by sum `09993148`), tokenizer v6 (`0640d3bd`), EOS after each document.

We computed the FineWeb-Edu 10BT budget from the actual counts of the remaining files, with a 0,06 bn margin for documents cut after the leakage scan. The cut turned out larger (2.1 % of EN, 9–10 % of legal), so the run has ca. 8 479 instead of 8 583 steps. We do not re-select data: the launcher computes steps from bytes, and every window is used once.

3.3 Why this composition

PL replay instead of EN alone. A larger EN share is admissible because there is simply less Polish text. A variant without replay would lower the core share in the whole diet (23.5 bn after 1.07) from 18.6 % to 11.5 %, and legal from 9.2 % to 5.7 %. This is a shift of the distribution toward EN, though much smaller than the full language switch studied in arXiv:2403.08763. Replay costs a second epoch on part of the data: after cleaning, 0,983 bn of core returns (36 % of its tokens from 1.06) and 0,912 bn of legal (68 %), while the rest of the corpus is seen once.

FineWeb-Edu as the EN backbone. It has the simplest license (ODC-By), the largest scale, and is selected for educational content. FineMath and open-web-math add mathematics, and DCLM in a subset adds web diversity. We take DCLM only within a 0,245 bn budget out of the available ca. 0,68 bn, so as not to shift EN beyond need.

What we do not take. Sources of unknown license (cc-news, starcoderdata, wildchat, ultrachat, tiny-textbooks, scientific-papers, open-subtitles) and synthetic texts from models with restrictive terms (rule R1). ARC-MIX is dropped because it contains augmented items related to

component	bn tok. (built)	source and notes
PL web, HPLT reserve	3,120	pool files untouched by 1.06; 2 448 525 docs, `a00c6921`; after test-leakage cut and PII masking: 3,110 (`2480ac94`)
PL core, replay	1,000	subset of those seen in 1.06, random by hash, minus exclusion list `ca61b7ff`; after leakage cut 0,983
PL legal, replay	1,004	0,704 (half of `09993148`, in full) + 0,300 (half of `f300328c`); after leakage cut 0,912 (long legal acts drop in full on a single 13-gram)
EN FineWeb-Edu (fresh from minimal-en-corpus-5b)	0,978	ODC-By; 840 133 docs; 36 996 excluded as copies of EN 1.06
EN FineMath	0,237	ODC-By; 152 623 docs
EN open-web-math	0,225	ODC-By; 86 621 docs
EN DCLM-baseline (subset)	0,245	CC-BY-4.0; 177 946 of 577 235 candidates, cut on whole documents
EN FineWeb-Edu `sample/10BT` (4 files)	2,254	ODC-By; 2 035 410 docs, `daa45a29`; 288 312 copies of EN 1.06 and 250 222 copies of rows excluded above
total	9,06	after leakage cut and masking: **8 891 005 707** tokens in 9 files (section 4)

Table 2: Composition of 1.07: `tokens_out` from the run manifests (with EOS). EN from minimal-en-corpus-5b in total 1 686 034 957 (`bfae642f`), after full cleaning v2: 1 650 618 814 (`332589ee`); FineWeb-Edu 10BT after full cleaning v2: 2 235 660 025 (`a1e95968`).

ARC, which would contaminate the ARC-Easy axis. F/K data from batch 1 were not included in the prepared pool of nine files. The result of the completed probe (section 7) does not trigger an automatic change of the dataset; the decision about additions belongs to the training team.

3.4 Leakage cut

Each file passes through `docs-bin`, and documents with a 13-gram hit drop out in `usun-bin` before training (section 4). The scan of the batch 1 base showed that this is not a formality: in the 105 mln tokens of the batch 1 base (a mixture of web, core, legal, and EN) there were 165 documents with fragments of reading-comprehension test passages (mainly PolQA).

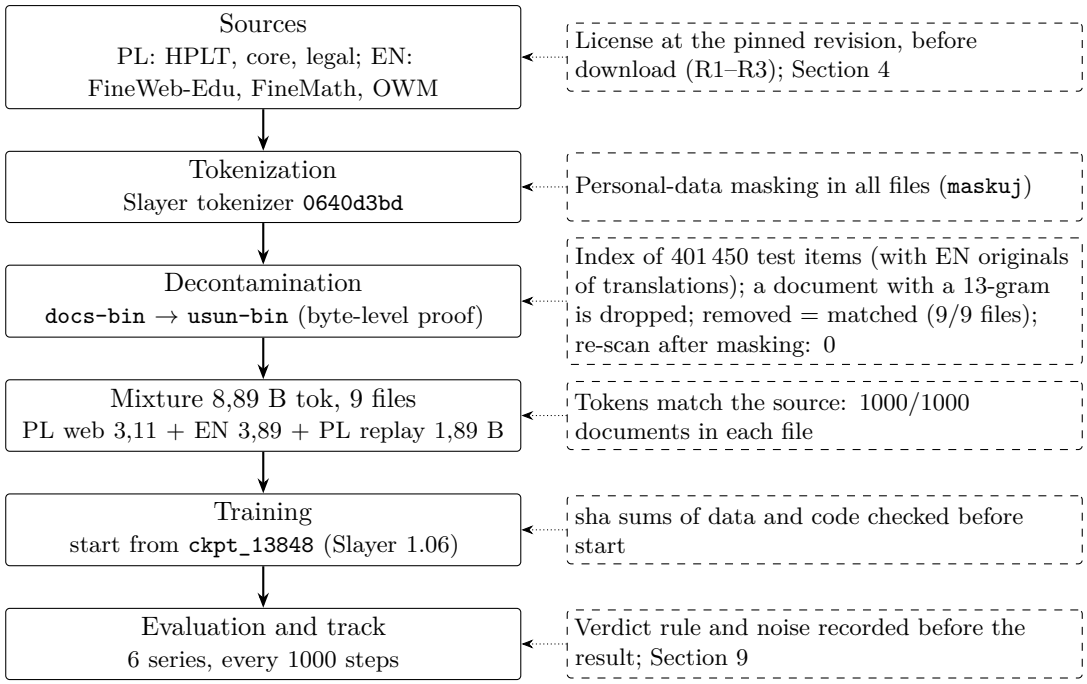


Figure 1: Slayer 1.07 data path. Stages on the left, gates on the right (dashed): data moves on only after a gate is met. After all gates, 8,89 billion tokens remain in 9 files; composition in Section 3, decontamination results in Section 4.

4 Data: licenses and decontamination

Each data source passes two checks before it reaches training: a license check before download and a scan for leakage of test items on the finished training file. Both produce a record with checksums, which goes into the model card.

4.1 Licenses

We admit a source only when the dataset's license at the pinned revision permits training and publication of the weights. Three rules reject sources regardless of the license of the dataset itself:

- **R1:** no texts generated by models whose terms forbid training other models on their outputs (GPT, Gemini, Llama 3.1);
- **R2:** data files only (parquet), without loading scripts and without executing code from the dataset repository;
- **R3:** no OpenStax content, including copies and adaptations on other platforms, because the publisher's terms forbid use for training language models.

Rule R3 turned out to be important in practice. In the LibreTexts dataset the input-set filter (the word „openstax" in the identifier, author, title, book address, or content) flagged 1 476 documents in neutral domains (mathematics, statistics) and 5 772 of ca. 13 thousand (ca. 44 %) in target domains (physics, chemistry, biology, medicine). All were cut before building the input sets.

4.2 Index of protected items

The scan compares the training data against an index of all test items by which the model will be measured: 263 255 items from the 50 tasks of our board (bench50) and 132 543 items from the

source	license	verdict
FineWeb-Edu	ODC-By 1.0	yes
FineMath	ODC-By	yes
open-web-math	ODC-By (publication)	yes, with a record of the license source
DCLM-baseline 1.0	CC BY 4.0	yes („research only" applies to the DCLM models)
HPLT 3.0 (PL)	CC0 of the compilation, TDM art. 4 DSM	yes
LibreTexts (without OpenStax)	CC BY / CC BY-SA / PD / GFDL per document	yes
Nemotron-CC v2	NVIDIA license (gated)	no, without reading the terms
GPT/Gemini/Llama synthetics	generator terms	no (R1)
OpenStax and derivatives	LLM-training ban	no (R3)

Table 3: License verdicts for the 1.07 sources.

datasets measured by the external replicating team (full BLiMP, MultiBLiMP-PL, WikiText-2, Belebele-PL, MMLU, Global-MMLU-PL, LLMzSzŁ, PIQA, SIQA, HellaSwag), each at the pinned revision. Together 395 798 items. After external review the index was supplemented with datasets that the training board measures but which were missing from it, and with the English originals of translated datasets: ARC-Easy (2 376), ARC-Easy-PL (2 376), and the English Belebele texts from FLORES (900), together 5 652 items. All 1.07 files were scanned on this index as well. The EWoK dataset is not in the index (a password-protected archive, the license restricts circulation), so we report results on it as unscanned.

4.3 Method

The text is lowercased and split into words (`\w+`). An item in which at least one content field has at least 13 words is represented by all the word 13-grams of its fields. When no field reaches 13 words, the item is represented by whole fields of length at least $\max(3,\ 0{,}6 \cdot L)$ words, where L is the length of the longest field (exact match). Categorical fields (labels, subject names) are omitted. A training document that contains even one item 13-gram is **removed** before training. Short matches (a short answer or an option present in the text) are counted and reported, but we do not remove documents on their basis, because they are not item leakage. Every hash hit is verified word for word against the item text. Every scan, including the scan of tokenized files, has a positive control (planted items, which in tokenized files are encoded with the same tokenizer and passed through the same path, must be found in full) and a negative control (random text must not produce a hit); failure of either halts the scan. For tokenized files the scan decodes the file with the tokenizer pinned by checksum and runs in parallel on all cores. Before scanning it checks that the file was produced with that tokenizer: for 100 documents spread across the file, re-encoding the decoded text must reproduce the same tokens in at least 95 % of cases (a file with a foreign vocabulary would give a falsely clean result; on PL data it comes out 100/100, and after a vocabulary permutation 0/100). Removal of documents is done on the same file, whose checksum, layout, and document count are checked against the scan report. The removal result is compared with the input document by document, byte for byte: every retained document must be identical to its corresponding input document. The scan result of each retained document is therefore the same as in the first pass, and a second full scan is not needed.

4.4 What the scan found before the first run

In the first version of the format input set (AQuA math tasks) 819 of 2 985 items of the MathQA test (27 %) had 13-grams of the content or the solution in the data. MathQA was built from AQuA, and the earlier filter compared only the questions. Removing 2 646 rows from three input sets gave 0 13-gram hits in a re-scan (short matches on answer options remained and are reported as a count). In the shared base (a mixture of the same family as the 1.06 data) 165 documents (0,6 % of tokens) contained fragments of Wikipedia on which DROP, PolQA, PoQuAD, SQuAD2, and BoolQ are built. A second sample of the same pool (the control arm, 77,5 % of shared documents) had 162 such documents in 14 of the same tasks. The target tasks of the first experiment had 0 13-gram hits in both samples; 15 tasks with a hit were excluded from the interpretation of this experiment. In 1.07 training such documents are removed without exceptions.

4.5 Result for the 1.07 data

file	documents removed	tokens removed	tokens after
EN: FineWeb-Edu, FineMath, open-web-math, DCLM	8 065	35,4 mln (2,10 %)	1 650,6 mln
EN: FineWeb-Edu `sample/10BT`	6 065	18,5 mln (0,82 %)	2 235,7 mln
PL: web (HPLT reserve)	2 924	9,6 mln (0,31 %)	3 110,4 mln
PL: core and legal replay (6 files)	17 454	109,6 mln (5,47 %)	1 894,3 mln
total	34 508	173,1 mln (1,91 %)	8 891,0 mln

Table 4: Documents with a test-item 13-gram removed from the 1.07 files before training.

In the English data the most hits concerned math tasks (MathQA), MMLU, and the English-language parts of the LLMzSzŁ exams; in the Polish web, mainly Wikipedia-based reading tasks. In replay, the most that was lost was legal texts (9–10 % of tokens): some test questions quote provisions, and a single shared 13-gram removes a whole, usually long, act. Hits in ARC-Challenge were few (12 and 3 documents). The supplementary index found in the English data a further 103 documents: 82 with FLORES texts (the source of Belebele) and 21 with ARC-Easy questions; in the Polish data it found nothing. All matched documents were removed without exceptions.

4.6 Limitations

The scan detects verbatim and near-verbatim leakage. It does not detect paraphrases or tasks from the same template family (example: AQuA and MathQA after removing shared 13-grams), so results on such tasks we describe with the label „source shared with the data”. This applies also to the PolQA and PoQuAD tasks: the passage corpora from which they were built are part of the Polish core used since 1.06 (also in the 1.07 replay). The scan removes documents containing an item 13-gram, but not the whole source corpus, so the results of these two tasks in 1.07 are only descriptive and do not enter the evaluation against 1.06.

Personal data (e-mail addresses, PESEL, NIP, REGON, ID-card, and phone numbers) are masked in every 1.07 file with the same tool. The Polish web comes from a pool that was cleaned of personal data already at preparation time (in 2,7 mln source documents one e-mail address remained, absent from the selected part); masking changed a further 4 426 documents in it (0,18 %), mainly phone numbers. Documents changed by masking are re-scanned.

5 Training procedure

The configuration below is a variant for review and approval by the training team, not a machine allocation nor a go-ahead to launch the run.

5.1 Start and step geometry

1.07 is a continued training of 1.06: it starts from checkpoint 13 848 (`1fc7396c`) together with the optimizer state (`--resume`). The trainer is r24.2 (`46f83b51`): r24.1 (`881cb055`, used to train 1.04) with an added optional diagnostic `--global-batch-sha`. Without this flag the values, data order and math are the same. 1.04–1.06 were trained on 4 and 8 L40S cards; the working variant of 1.07 assumes 8 RTX 4090 cards (or 5090). The step geometry is the same: $2 \times 16 \times 8 \times 4096 = 1\,048\,576$ tokens (micro-batch × accumulation × cards × context). With 8 891 005 707 tokens after v2 cleaning this gives about 8 479 steps, i.e. ending around step 22 327; the exact count is computed by the launcher from the bytes of the concatenated file.

5.2 Learning rate

The LR is constant: 6e-4 (AdamW) and 0,02 (Muon), without decay (`--decay-frac 0`), as it has been since 1.04. A constant LR keeps the path open to further stages (goal 40 billion) without re-warming. In a 150-step experiment from 13 848 the loss change relative to 1.06 was −0,0503911 nat for decay (A) and −0,0034372 nat for constant LR (B). The A−B contrast is −0,0469539 nat (SE 0,0006), not the change of A relative to 1.06 (repository benchmarks-fabryka-ai, `research/slayer-1b/wygaszanie`). According to arXiv:2405.18392 the gain grows up to about 20 % of steps, and weight averaging does not match it. Therefore at the end of the stage we plan a **release branch with decay** (about 850 steps from the last constant-LR checkpoint, 13–20 $), compared against avg10. The main run stays on constant LR, so that the next stage starts without re-warming.

5.3 Data order

The trainer reads a single `train.bin` file. We therefore concatenate the component files (table 2) into one file, and the epoch sampler draws a permutation of windows across the whole file. The sources thus mix proportionally to the token count, and each window is used at most once. The stream counter starts from zero (`--data-offset-steps=-13848`), so 1.07 does not skip any part of the new file.

Arranging data in ascending order by quality under constant LR (arXiv:2511.18903, section 6) does not enter 1.07. In our trainer the window order is set by a permutation, not by the file layout, so such a change requires a new sampler, i.e. new code before the gate. On top of that, a quality score (`score`) is available only for part of FineWeb-Edu. This is a candidate for a cheap probe before the next stage.

5.4 Final phase

The prepared 1.07 set does not contain the F/K data from batch 1. Reading the completed probe separates the BPB result from the domain-specificity hypothesis (section 7). Any inclusion of the data requires a separate team decision; training phasing alone does not replace data qualification.

5.5 Checkpoints and avg10

The trainer saves state every 250 steps in case of failure, and the launcher keeps copies of it (a hardlink with a `sha256` sum) on a grid:

- every 1000 steps (14 000 … 22 000),
- milestones 13 998 (control, below), 15 730, 18 590, 21 450 (extension of the 1.04 grid),
- the last 10 checkpoints every 50 steps, including the last one.

From the last ten an averaged model (avg10) is produced. For release we pick the checkpoint according to heldout PL (section 9). The comparison baseline for 1.07 avg10 is 1.06 avg10, not the single checkpoint 13 848, because averaging the 1.06 weights alone gives about −0,045 nat on heldout.

5.6 Control: the same steps on 1.06 data

Between 1.06 and 1.07 several things change at once (the EN share, new PL web, a second epoch of core and legal, the step count). To separate the effect of data from the effect of continued training itself, we have a control arm that already exists: arm B of the decay experiment. These are 150 steps from 13 848 on a constant LR 6e-4 on a mixture of the same family as 1.06 (the `slayer-pl-8x3b` set at revision `d50df04a`). Therefore 1.07 saves a checkpoint exactly at step 13 998, and we read the effect of data at the start of the run as $\Delta = 1.07@13\,998 - B@13\,998$, on the same axes. For longer segments the null hypothesis "longer training" is the 1.04→1.06 trend recorded before the start (section 9).

5.7 Pre-start checks

The launcher stops if the `sha256` sum of the trainer, tokenizer, checkpoint 13 848 or any data file does not match the one entered. After the start it checks in the log for `RESUME @ 13848`, the parameter count and the card count. Before training: 10 minutes of throughput measurement on 8 cards and an NCCL link test (on vast machines with 8 RTX 4090 cards `NCCL_P2P_DISABLE=1` is required).

6 Training methods: what is proven and what is a hypothesis

We adopt a method when it has evidence in the literature at a scale close to ours or in our own measurement. The numbers below are results of the authors of the works, not our measurements. Transferring each of them to Slayer (1B, PL+EN, continued training from 13 848) is a hypothesis to be tested according to section 9.

6.1 Arranging data over time

Ascending order by quality under constant LR (candidate, not in 1.07). Luo et al. (arXiv:2511.18903, ICLR 2026) train 1,5B models on 30 billion tokens. Arranging data in ascending order by quality *clearly* wins for them over random shuffling under constant LR, but the advantage disappears under standard LR decay. Moderate decay together with averaging the last checkpoints gives +1,64 % on average on a benchmark suite. Slayer trains on a constant LR $6 \cdot 10^{-4}$ without decay, and averaging the last checkpoints (avg10) is in our plan. Decay we have already measured: after 150 steps the loss change relative to 1.06 was −0,0504 nat for decay and −0,0034 nat for constant LR. The contrast decay minus constant LR was −0,0470 nat (SE 0,0006; section 5). Hägele et al. (arXiv:2405.18392) show that a constant LR with a short decay scales like a cosine schedule, and weight averaging improves the result along the run at no extra cost. A quality label for EN is in the data (FineWeb-Edu: columns `score` and `int_score`), but only in the FineWeb-Edu part. **In 1.07 we do not use this method:** the trainer's sampler draws a permutation of windows across the whole file, so the order of documents in the file does not carry over to the training order. It would require a sequential sampler, i.e. new trainer code

(section 5). **Status: a hypothesis based on a single work at a close scale,** a candidate for a separate trial under constant LR.

Final phase with domain data (candidate). Blakeney et al. (arXiv:2406.03476) boost domain data in the last 10–20 % of training of a 7B model (1 trillion tokens) and obtain up to +6,90 pp on MMLU, +8,26 pp on GSM8K and +6,17 pp on HumanEval over the base mixture. For us this is the place for format and knowledge data from the 1.07b probe, if the probe shows movement above noise (section 7). **Status: a hypothesis; the work's scale is about 7 times larger in parameters.**

Language curriculum from easy to hard (rejected). Campos found no convincing evidence that a curriculum improves training of a language model (evaluation on GLUE; arXiv:2108.02170). The summary of BabyLM 2023 judges curriculum attempts as „largely unsuccessful" (Warstadt et al., CoNLL 2023, ACL Anthology). We do not use it.

6.2 Distribution shift: more EN after training mostly in Polish

Ibrahim et al. (arXiv:2403.08763) study continued training under a language change (EN→DE, 405M). According to the abstract, to match training from scratch on all data, a combination of three things *suffices*: replaying part of the old data, re-raising the LR and re-decaying it. Gupta et al. (arXiv:2308.04014, Pythia 410M) show that re-raising the LR first increases the loss on old and new data, and in the long run improves the result on new data.

The conclusion for 1.07 is weaker than in those works. They study a full language change, and for us Polish is still about 70 % of the diet after 1.07. **We therefore treat a constant PL share (replay of core, legal and PL web) as caution, not a requirement following from those works.** The PL axes count as a cost in the verdict rule (section 9). Re-raising the LR does not apply to us, because we train on a constant LR. It would only apply with a schedule change.

6.3 Instructional text and solved exercises

Gunasekar et al., arXiv:2306.11644v1, study the Phi-1 model of 1,3 billion parameters on a Python code generation task. Their pretraining set has fewer than 7 billion tokens: about 6 billion of filtered code and fewer than 1 billion of synthetic textbooks. The model nevertheless saw **more than 50 billion tokens**, because the data were repeated. The size of the set is not the number of the model's exposures. After pretraining comes fine-tuning on about 180 million tokens of exercises. The authors report about 29 % HumanEval for Phi-1-base and 50,6 % for the model after the CodeExercises stage. We do not attribute this difference to the textbooks alone: the training stage and the added exercises also change.

What is transferable is the selection principle: a self-contained description, an example, a correct solution and varied exercises instead of illegible fragments or repeated boilerplate. What is not automatically transferable is the numerical result: the work concerns code, not a general PL+EN model. The textbooks and exercises were generated by GPT-3.5, so **we do not import their data under rule R1**. Adaptation on permitted texts and programmatically checkable tasks remains a hypothesis to investigate.

Support for quality selection is SmolLM2, arXiv:2502.02737v1. The authors compare web sources at an equal model and budget of 350 billion tokens; FineWeb-Edu and DCLM give different strengths. This justifies choosing the mixture for skills, rather than copying a single ratio of sources as universal. The full 1,7B model was trained on about 11 trillion exposures. Its stages change data, LR and token count simultaneously, so comparing them does not isolate the causal effect of order alone.

6.4 Interleaving text with tasks in pretraining

Cheng et al., arXiv:2406.14491v2, propose *instruction pre-training*: to part of the texts they add tasks based on their content. The training objective stays the same as in ordinary pretraining: next-token prediction and loss on all tokens. This is neither RLHF nor fine-tuning after pretraining alone. In a comparison of 1,3B models trained on 100 billion tokens at an equal budget the result depends on the axis:

Metric	Ordinary pretraining	Instruction pre-training
ARC-C	28,8	30,9
MMLU	25,7	27,3
HellaSwag	54,9	54,3

This is not proof of improvement on every skill. In the work the pairs were created by a fine-tuned Mistral-7B synthesizer; the generator's conditions, the provenance of its data and leakage require separate control before using such a corpus. A close parameter count does not remove the difference between pretraining from scratch on 100 billion and a short continued training of Slayer.

6.5 A hypothesis for the second batch, without changing the 1.07 run

Hypothesis: with the same pool of facts and budget, interleaving a description with programmatically checkable questions improves the use of information more than text alone, without an unacceptable cost to Polish. We propose fresh C, C2 and I from the same 1.06 checkpoint `ckpt_13848`, pinned in the batch-one code package. C/C2 get texts from the same pool as I, but I replaces the proposed 20 % of domain slots with text interleaved with tasks. The answers follow from the text or from a computational rule; no external LLM and no test sources. The 20 % share is a proposal parameter, not an optimum shown in the work.

The tokens of the whole arm, the remaining base, the PL replay, the packing, the LR, the optimizer and the checkpoint choice stay the same. One must check and report the number of fact exposures: an additional repetition of a fact in Q/A must not be an unnoticed second factor. In this trial we choose variant (a): C2 contains exactly the same documents and the same number of exposures as C, but in a different permutation; we check equality of the multiset of documents. Thus C2 − C measures order noise, not sampling selection. We do not copy the way C2 was built in batch one, where a different sample of the pool arose. We also do not use its ready controls, because this trial has a different mixture.

The proposal's budget is 100 steps per arm at 1 048 576 tokens per step, i.e. 314 572 800 processed tokens in total. The price and throughput must be measured before the run. Before and after we measure the same held-out tasks, also in a changed layout, cloze/BPB of the answers and the mandatory heldout PL. Before the result we record the main axis, the minimal useful effect, the threshold against the C2 − C noise and the acceptable PL cost. The trial may remain unresolved because of the small dose; a lack of gain does not refute the result of the work on 100 billion tokens. This is a controlled-experiment design, **not a performed measurement** and not a condition halting the preparation of the 9 billion for the 1.07 training.

7 Probe 1.07b (batch 1): do format and knowledge move the result

Question. Does a small admixture of format data (F) or domain text (K) move the targets from section 2 above training noise, before we spend the budget on full 1.07?

Design. Four arms start from 1.06@13848: one epoch, 100 steps of 1 048 576 tokens (104 857 600 training tokens per arm), linear LR decay to zero:

- **C**: baseline (scenario B mixture from the benchmark team's measurements);
- **C2**: a different sample of documents from the same pool, in a different order (a different sort key before truncation). The difference C2 − C therefore covers both ordering noise and data-sample noise;
- **F**: C with the slots swapped to format data: aqua_rat (48 882 items) and synthetic PL/EN fill-the-gap questions without an LLM (100 056 items), templates outside the evaluation;
- **K**: C with the same slots swapped to domain textbooks (LibreTexts EN, AGH chemistry PL; 9 229 items), without questions.

The insert has 13 891 979 tokens in both F and K and occupies the same positions in the file. Mixer acceptance: F enters at 100 % in every source and language, and K at 99,89 % (EN) and 99,77 % (PL).

Reading rule (recorded before the results). For a task and metric $\Delta = \bar{m}^{\,\mathrm{arm}} - \bar{m}^{\,C}$ on shared positions, SE(Δ) from the paired bootstrap ($B = 10\,000$), $z_k = |\bar{m}^{\,C2} - \bar{m}^{\,C}|$. A move occurs when $|\Delta| > 2\,\mathrm{SE}$ and $|\Delta| > 2\,z_k$, and at $|\Delta| < 3\,z_k$ we call it fragile. Success of F requires an upward move of letter accuracy on the full set plus condition (a): the lower bound of CI95 above the maximum of the random and majority levels, on the panel for MMLU/MedMCQA, on the full set for LLMzSzŁ. The alternative is an upward move of accuracy on the format probe. In both cases the dominant-letter share must be $< 50\,\%$; a drop of that share alone is not enough. The recorded success rule for the knowledge metric is a downward move of correct-answer BPB. Separately, K specificity was predicted: a move on the TARGET part of MMLU without such a move on the CONTROL part. Improvement on control we report as transfer; it does not invalidate the success of the BPB metric. The condition $|\Delta| > 2\,z_k$ is necessary here, not precautionary. In the negative control on the benchmark team's data (150 steps of constant LR on 1.06 data, with no new data; section 9) correct-answer BPB shifts on all 8 tasks by $z = 12$–62 computed with the paired SE alone, while acc_norm does not shift on any cloze task. The paired SE alone measures question-sample noise at fixed weights, not the movement of weights between checkpoints. The heldout used in the batch 1 evaluation (`val.bin`, the same as in the decay experiment) overlaps with the arms' training data, so we report its BPB descriptively only.

Decontamination and exclusions. The F and K input sets passed a 13-gram scan on an index of 395 798 test items with 0 hits (MathQA: 819 items before the cut). Baseline C has hits in 165 documents across 15 tasks, mainly from Wikipedia passages (DROP, PolQA, PoQuAD, SQuAD2, BoolQ). Of these, 18 documents lie in the slots swapped in F and K, and C2 is a different sample of the pool. The leakage is therefore asymmetric across all arms, so these 15 tasks do not enter the conclusions. The targets (ARC-C, Belebele-PL, MMLU, LLMzSzŁ, MedMCQA) have 0 hits. MathQA is excluded additionally, because AQuA and MathQA share a source.

Main result. We read the completed C/C2/F/K and the reference point 13 848 from the final results package, without the drawer. The thresholds recorded before training were kept. The completeness acceptance covered 69 files. An independent implementation confirmed, on 10 contrasts, the denominators, Δ, z_k, the direction of the move and fragility; small SE differences from bootstrap sampling did not change any classification.

Table 5: Format F versus C: letter accuracy and the necessary condition of no letter dominance.

task	n	Δ (pts)	2 SE	$2z_k$	dominant letter F
MMLU	14 042	−0,021	0,570	0,541	A 77,56 %
MedMCQA	4 183	−1,960	0,966	0,574	A 76,98 %
LLMzSzŁ	18 815	−0,096	0,371	0,117	B 92,90 %
format probe	432	+0,231	4,668	0,926	A 51,39 %

F has no upward move of accuracy on any target or probe; on MedMCQA there is a downward move. The condition of dominant-letter share $< 50\,\%$ is not met on any of these axes. In the shifted layout the accuracy of the targets stays in the noise. Cost on the Belebele cloze control: −2,00 pts ($2\,\mathrm{SE} = 1{,}56$, $2z_k = 0{,}22$); Belebele from letters stays in the noise.

Table 6: Knowledge K versus C: correct-answer BPB; lower is better.

task	n	Δ BPB	2 SE	$2z_k$	move
ARC-C cloze	1 172	−0,042139	0,009884	0,001949	improvement
MedMCQA cloze	4 183	−0,038371	0,004562	0,032080	fragile improvement
MMLU TARGET	3 037	−0,034209	0,004375	0,001714	improvement
MMLU CONTROL	6 219	−0,003598	0,001366	0,001124	control improvement

The split of MMLU into 16 TARGET subjects, 24 CONTROL and 17 neutral was recorded before corpus selection. K improves TARGET BPB more strongly than CONTROL, but both groups move in the same direction. This does not confirm the predicted specificity, but it does satisfy the BPB-move rule; we do not turn the prediction into a new success criterion after the result. K accuracy on the seven full tasks stays in the noise. In the fixed MMLU TARGET subgroup, however, it rises by 1,087 pts ($2\,\mathrm{SE} = 0{,}884$, $2z_k = 0{,}395$); the rise of 0,418 pts on CONTROL stays in the noise. This is an auxiliary result relative to the registered BPB metric, not a new criterion.

Decision and scope. F does not meet the format criterion. K has success of the BPB metric, but does not confirm the domain-specificity hypothesis. The prepared pool of nine files does not change automatically; the decision to include the additions belongs to the training team. This is a result for this dose and this design, not proof that a larger or different insert will not work. The descriptive change of C relative to 1.06 is −0,038174 BPB on ARC-C and −0,009919 on MMLU 5-shot; we do not add it to F − C or K − C. A full BLiMP control is missing in this package, and the retention heldout is not an independent measurement of generalization.

8 Audit of the thematic composition of the data

Why. The data selection described in section 3 rests on source, license and educational score (FineWeb-Edu), and not on a per-document check that the content overlaps with the knowledge gaps measured in section 2. This audit is an approximate, reproducible attempt at such a check: it shows how a sample of documents distributes over domains and whether the new files differ from the 1.06 replay.

Method. From each file we take 700 documents evenly along its length (a fixed selection, a deterministic result), decode the first ∼220 tokens with the Slayer tokenizer and compute an embedding with the model `intfloat/multilingual-e5-base` (offline, no network). Each document we assign to the nearest of seven domains by cosine similarity to a hand-written

domain description. **This is zero-shot classification with hand-written descriptions: the exact percentage values have an error on the order of a few points, and the ordering of files is more reliable than a single number.** Code and result: `audyt/audyt_skladu.py`, `audyt/audyt-skladu.json`.

Table 7: Share of the target domains (natural sciences and medicine + mathematics and computer science) in the sample. The TARGET column is their sum; next the largest off-target component.

file	group	TARGET	sci./med.	math./cs.	largest off-target
`en_mix`	new	**57,9 %**	36,3 %	21,6 %	web 13 %, history 12 %
`en_fineweb_edu`	new	41,7 %	31,9 %	9,9 %	web 20 %, history 16 %
`pl_web`	new	37,6 %	27,4 %	10,1 %	web 43 %
`pl_replay` core	1.06	32,1 %	25,6 %	6,6 %	web 53 %
`pl_replay` legal	1.06	25,9 %	14,3 %	11,6 %	legal 41 %

Reading. The new English data are more clearly concentrated on the target domains than the 1.06 replay: `en_mix` (a mixture of FineWeb-Edu, FineMath, open-web-math and DCLM) has about 58 % science/medicine/mathematics content, mainly through 22 % mathematics and computer science, versus 32 % in the replay core and 26 % in the replay legal. This is a measured enrichment with knowledge that 1.06 had little of.

What this does not say. Even the best file is about 42 % off-target content (web, history, social sciences), and `en_fineweb_edu` is below half on topic. `pl_web` is mostly general Polish web, not domain knowledge. **The knowledge enrichment concerns almost exclusively English**; of targeted Polish domain material this set adds practically nothing, and it is precisely Polish textbook knowledge that was one of the measured gaps of 1.06 (section 2). The assignments of general Polish web to "science/medicine" (27 % in `pl_web`) are probably inflated by the classifier, so we read them cautiously. The audit is a signal about composition, not a measurement of effectiveness: whether this shift moves Slayer's weak points is decided only by the post-training measurement against the control arm (section 9), and the batch 1 probe (section 7) signaled that the BPB move alone was not domain-specific.

9 Measurement: when 1.07 is better than 1.06

9.1 Two levels of measurement

We measure 1.07 at two levels, each with different code and for a different purpose.

1. **Training timeline (track).** The same board as for 1.04–1.06, computed on successive checkpoints and published on track.fabryka.ai. It serves to follow the run and for the EN axis fuse.
2. **Full 1.07 card axes.** Full test sets with a result for each position, computed on checkpoints from the cadence. From them comes the verdict "better than 1.06" (the verdict rule in this section).

We do not combine the results of the two levels. BLiMP in the track protocol (76,12 for 1.06) and BLiMP in the benchmark team's protocol (79,21) are different protocols. We compare only 1.07 with 1.06 in the same protocol.

9.2 Track protocol (v8, for 1.07 as v9)

Code: `evale_auto_v8.py`. The v8 → v9 change concerns only where the checkpoints are taken from, and the new 1.07 run. The way of computing does not change.

- **EN:** ARC-Easy (2 376 questions, accuracy), BLiMP (67 000 pairs, accuracy), WikiText-2 as byte-PPL = $\mathrm{PPL}_{\mathrm{tok}}^{n_{\mathrm{tok}}/n_{\mathrm{bajt}}}$ from the unrounded token PPL (the token PPL remains as diagnostics).
- **PL:** MultiBLiMP-pl (3 272 pairs) and ARC-Easy-PL EU20 (2 376 questions).
- **Calibration before the first point:** Slayer1B-1.02 from HF (revision `bf7117b9`) must give exactly the numbers from the 1.02 card: ARC-Easy 48,99, BLiMP 74,71, byte-PPL 2,3735, MultiBLiMP-pl 3 219/3 272, ARC-Easy-PL 994/2 376. A different number stops the publication.
- **Time:** about 989 s per point on a single A100 (measured on 28.10). We did not measure the time on the 4090.
- **Where 1.07 is computed:** a separate 1×4090 card in the same region as the training machine. The checkpoint is pulled directly from the training node by `rsync` read-only. A one-time key restricted to the address and directory, which disappears together with the machines, is used for this. Before the computation the sha of each checkpoint is compared with the trainer's `SHA256SUMS`. The result is sent to the track from the station. The first publication of the run's name and label passes a content review (section 4).

1.06 as the starting point. The 1.06 run on the track (`f056bb5c`) has both end-of-run checkpoints and 22 constant-LR points at steps 10 010–13 848. The difference between the two last checkpoints (z_k, definition in the verdict rule in this section) is only one pair. Alongside we give the standard deviation of the residuals around the linear trend on these 22 points (SD of residuals, 20 degrees of freedom):

series	@13800	@13848	z_k	SD of residuals
BLiMP (%)	75,82	76,12	0,30	0,45
ARC-Easy (%)	52,15	52,99	0,84	0,58
WikiText-2 byte-PPL	2,2882	2,2849	0,0033	0,0046
MultiBLiMP-pl (%)	98,50	98,41	0,09	0,15
ARC-Easy-PL EU20 (%)	45,20	45,03	0,17	0,40

Columns: the 1.06 result at steps 13 800 and 13 848 (accuracy in %), z_k from this pair and the SD of residuals from 22 points. On four of the five axes the pair understates the noise: on MultiBLiMP-pl and ARC-Easy-PL EU20 adjacent points among these 22 differ by more than $2z_k$ in 52 % of pairs. On ARC-Easy it is the opposite. Both numbers come from a single run, so they do not include seed noise or data-order noise. The SD of residuals also contains the real training fluctuations between checkpoints, not just measurement noise. Which quantity the verdict rule takes is set by the rule in this section.

9.3 Full 1.07 card axes: how we compute them

The axes and metrics are set by the 1.07 card. Here we describe only the code and the correctness proofs.

- **Scorer:** the log-likelihood of each option, computed by the benchmark team's scorer (benchmarks-fabryka-ai repository; `hipoteza_formatu.py`, called by `ewal_ramion.py`, commit `4c2a6d6`, code unchanged). Tasks not present there are added by `diag107b.py`: letters A–D on MMLU (14 042), MedMCQA (4 183) and LLMzSzŁ, the shifted layout, the

format probe and MedMCQA in a cloze version. Each set file is pinned by its sha256 sum (`zbiory_pins.json`), and a mismatch stops the run.

- **Result:** one row per position (log-likelihood of each option, bytes, gold). From it we compute accuracy, BPB of the correct answer and the share of each of the letters A–D. We form 1.07 − 1.06 pairs by position identifier.
- **LLMzSzŁ:** 6 of 18 821 positions have 5–6 answers. In the A–D letter variant we skip them (n = 18 815), and the cloze − letters gap is computed on the shared identifiers.
- **Letters in one model pass.** The benchmark team's scorer computes each letter by a separate pass of the same context. Since a letter is one token, one pass per position suffices, and the cost of the letter tasks drops about 4 times. Condition of use: on a sample of 200 positions from each task $\max|\Delta \log p| \leq 10^{-4}$ and 100 % matching choices. Results:
  - tiny/CPU: $\max|\Delta| = 9{,}5 \cdot 10^{-7}$, 200/200 matching choices in each of the 7 tasks;
  - Slayer 1.06 @13848 on GPU: $\max|\Delta| \leq 2{,}4 \cdot 10^{-5}$, 200/200 in 7/7 tasks;
  - control: a version with a deliberately shifted read position gives $\max|\Delta| = 0{,}151$ and 0/50 matching choices, so the test distinguishes wrong code from correct code.

  In tokenizer 0640d3bd each letter after the context is a single token in all positions of the full sets, so the variant applies everywhere. We record how each task is computed in `wariant.json` next to the results.

9.4 Cadence

- A full track board every 1 000 steps (about 1 billion tokens), plus mandatory points: the start 13 848 (= 1.06), the end of each data phase and the final checkpoint.
- The last 10 checkpoints every 50 steps (for the avg10 average) we compute with the heldout loss alone, without the full board.
- We do not compute more densely: the BLiMP slope in the 1.06 run is unestablished ($p = 0{,}18$, and the swing between steps 8 600 and 10 010 is −0,59 pt; benchmarks-fabryka-ai repository), and the spread of the BLiMP residuals around the trend on the track is 0,45 pt. Points less than 200 steps apart therefore will not separate the trend from the checkpoint noise.

9.5 Result condition: leakage scan

A number from a task whose test positions have a 13-gram in the training data we report only with a leakage label and next to the result on positions without hits. This is how the 1.06 results page already works (scan of the 1.06 corpus, receipt `bc6c168c`). For 1.07 each data file passes, before training, a scan and removal of documents with a hit (section 4), so the card axes are meant to be computed on data with zero 13-gram hits. EWoK is not in the scan index, so its result stays without this guarantee.

9.6 Verdict rule (recorded before the run)

The question that 1.07 can settle. Between 1.06 and 1.07 the mixture, the EN share, the second epoch of the core and legal data, the number of steps and the hardware all change at once. The full run therefore yields at most the verdict "better than 1.06 after about 8.5k steps on the new mixture". We isolate the effect of *data* only over the segment 13 848→13 998, against the control arm B (section 5): $\Delta_{\text{dane}} = 1.07@13\,998 - \text{B}$, computed by code and on the tasks of the LR-decay experiment, for which the results of B are recorded per question.

Negative control of the rule. Before adopting the metrics, we checked them on the placebo pair B − 1.06: 150 steps of constant LR on 1.06 data, with no new data. The axis that "moves" here does not distinguish the effect of data from mere further training. Tab. 8 gives z computed from the pair SE alone.

Table 8: Negative control: $z = |\Delta|/\mathrm{SE}$ for B − 1.06 (no new data).

	cloze				letters			
metric	ARC-C	MedMCQA	SIQA	MultiBLiMP	ARC-C	MedMCQA	SIQA	PubMedQA
acc_norm	0,9	0,0	0,3	1,7	0,4	0,4	0,7	5,1
BPB of correct	15,7	25,3	11,9	32,1	24,8	22,0	61,9	14,3

The BPB of the correct answer shifts after 150 steps on the same data by 0,04 to 0,61 bits per byte (absolute value). The standard error of the pairs measures only the sampling noise of the question set at fixed weights, not the movement of weights between checkpoints. For this reason, in 1.07 BPB of the correct answer is a descriptive axis, while the verdict stands on acc_norm in the cloze format, pair accuracy and continuous-text BPB. Letter accuracy is also descriptive: on MedMCQA, 1.06 picks "A" in 93 % of questions, so a change in the favoured letter changes accuracy without a change in knowledge.

Difference and noise. For a task over positions $i = 1, \ldots, n$ shared by both models

$$\Delta = \frac{1}{n}\sum_{i=1}^{n}\left(m_i^{1.07} - m_i^{1.06}\right), \qquad \sigma = \sqrt{\mathrm{SE}(\Delta)^2 + z_k^2}, \qquad z = \Delta/\sigma.$$

We compute SE(Δ) by paired bootstrap over positions ($B = 10\,000$, seed 20261028), clustered by paradigms (BLiMP, MultiBLiMP-PL) and subjects (MMLU). The checkpoint noise z_k is the standard deviation of the residuals around the line through 10 checkpoints of 1.06 (13 400–13 848 every 50, the same ones from which avg10 of 1.06 was built), computed by the same code before the first reading of 1.07. On the track axes we take the larger of this value and the SD of the residuals from the 22 track points (protocol above). The difference of two adjacent checkpoints, which we used earlier, understates this noise.

Table 9: Main axis and the family of 1.07 verdict axes (locked before the run).

axis	metric
main: heldout PL outside 1.06 and 1.07 data	BPB ↓
ARC-Easy, ARC-C, MMLU, MedMCQA (EN)	cloze acc_norm ↑
BLiMP (EN)	pair accuracy ↑
WikiText-2 (EN)	BPB ↓
MultiBLiMP-PL (PL)	pair accuracy ↑
ARC-Easy-PL EU20 (PL)	cloze acc_norm ↑
descriptive: letters (MMLU, MedMCQA, Belebele-PL, LLMzSzŁ), BPB of correct, PolQA, PoQuAD	

Heldout. H107 was built from the HPLT reserve after excluding the identifiers of packs 0–7 and the p08 data (seed 1107). The split was done over whole documents, even and odd in a fixed order, not over consecutive windows:

part	documents	tokens with EOS	SHA256 of the binary file (prefix)
selection	1 056	1 042 856	`a1f904d5`
verdict	1 056	1 058 519	`d3042fee`

The files, indices and id lists are consistent with the H107 manifest checksums; the id lists of the two parts are disjoint. File integrity alone does not prove that the content is disjoint from training. A test against the 9 files of 1.07 v2 gave 0/2 112 identical full documents and 0/2 112 shared 64-token prefixes. A separate scan, however, showed at least one shared 13-gram in 681 of 2 112 H107 documents (32,24 %): 345 in the selection part and 336 in the verdict part. This is not the fraction of shared tokens. The examples include shared templates, but it was not shown that they explain all hits. The absence of full copies is not proof of the absence of near copies. Copies of content from packs 0–7 were not checked. A second heldout H' outside HPLT (FineWeb2-HQ `pol_Latn`) has been built: 1 561 documents, 1 050 043 tokens with EOS, `H2.bin` `de8030e6` (manifest `ea431d93`). Against each of the 9 v2 files no identical full document (0/1 561) nor 64-token prefix was found (0/1 560; one document is shorter than 64 tokens without EOS). The 13-gram scan, however, gave hits in 256/1 561 documents of H' (16,40 %); this is neither the share of shared tokens nor proof of memorisation. The control of the 1.06 packs remains not done. The earlier heldout from the LR-decay experiment is descriptive. Its core and legal part lies in the 1.07 replay (33 % and 70 % of documents identical in full), and its EN and web part has no identical documents in the 1.07 data, but 1.06 saw it. It therefore measures retention, not the gain on new text. The first part of H107 serves only to select the checkpoint, the second only for the verdict (a). All windows of a single document stay in the same part. This limits the reuse of the selection data; we do not equate document disjointness with statistical independence.

Sensitivity to shared fragments. The main axis and the checkpoint selection stay on the full H107. Before reading the models, an additional analysis was recorded without documents with any 13-gram hit against the 9 files of 1.07: 711 selection documents remain (582 307 tokens with EOS) and 720 verdict documents (629 809 tokens with EOS). The split and the order are preserved. For a once-selected checkpoint we will compare paired ΔBPB on the full verdict part, its documents without hits, and the documents with hits, without re-selecting the checkpoint. BPB is the sum of losses in bits / the sum of bytes; we run the bootstrap over whole documents. A sign reversal after discarding the hits will be reported as a dependence of the result on coverage. We do not call the part without hits semantically clean or disjoint from the 1.06 data. The same descriptive analysis will cover H': after discarding the 256 documents with hits 1 305 documents and 747 491 tokens with EOS remain. We report the full H' as well as the parts without hits and with hits for the same checkpoint, without re-selection. The full H' remains the control in condition (a); the subset does not replace it.

Comparison base. We compare avg10 of 1.07 with avg10 of 1.06, and the single checkpoint with 1.06@13 848. avg10 of 1.06 alone lowers the heldout loss by about 0,045 nat and raises BLiMP by about 1 pt, so comparing avg10 with a single checkpoint would attribute the averaging effect to the data.

Selection of the checkpoint for release. The only criterion is the lowest BPB on the half of the heldout assigned to selection. The form is chosen the same way: a single checkpoint or avg10.

The verdict "better than 1.06". On the chosen checkpoint, computed once, it requires simultaneously:

(a) a drop of BPB on the verdict half of the PL heldout, $z < -2$, and no rise on the second heldout;

(b) an upward move on at least one EN axis and one PL axis from the family in tab. 9, by the Holm procedure at $\alpha = 0{,}05$ for 8 axes;

(c) no downward move by the same procedure on any axis of the family.

A downward move on axis X changes the verdict to "better with a cost on X". Without (a) the verdict reads at most "different". Without correction a false "cost" in (c) with 8 axes and the threshold $z < -2$ would appear in about 1 in 6 runs. We do not average over tasks and do not build a ranking (arXiv:2203.08242). The axes are not chosen by the results. A new axis may be added only as descriptive.

Power. The minimum detectable difference at power 80 % and the sharpest Holm step is $3{,}57\,\sigma$. The pair SE alone already gives at least about 3 pt on ARC-C ($n = 1\,172$) and 1,6 pt on MedMCQA. We publish the table for all axes, with z_k, before the start. A smaller difference without a move means no resolution, not "no change".

Axis fuse. A drop of an EN family axis ($z < -2$, without correction) at a cadence point is noted within the same hour. We do not stop the run automatically; the decision to continue it is taken separately.

10 Cost, machines and data logistics

Sources of numbers: [R] vast.ai market read 01.11, 18:16 UTC (sha ea0cccb7), [S] account guardian log of the batch 1 instance up to 23:43 UTC (245798bf), [K] full log of that guardian up to instance deletion (6175ed03), [T] training log of arm C (72be455d), [N] project notes from 01.11. The source files are in the project's internal repository, not in the benchmark repository nor on HF. The [estimate] mark is a calculation, not a measurement.

Measured rate. Slayer 1B (1 073,9 M parameters, step 1 048 576 tokens) trains on a single RTX 4090 card at a rate of **12 286–12 299 tok/s**, that is about 85,3 s per step [T]. The measurement comes from batch 1: one arm on one card, without distributed training. The leakage scan run alongside on the CPUs of the same machine did not change the rate (12 297 tok/s during the scan, 12 300 before it [N]). An independent measurement by the benchmark team on another 1×4090 machine (the LR-decay experiment, `research/slayer-1b/wygaszanie`) gives **11 435–11 440 tok/s** (91,7 s per step). We did not measure the rate on 8 cards (DDP). For the calculation we take a scaling efficiency of 0,88 [estimate], that is about 86,5 thousand tok/s on an 8×4090 machine from our measurement; at the benchmark team's rate it comes out to about 80,5 thousand tok/s, and the stage time is about 7 % longer (1.07: about 31 h instead of 29 h).

Prices (vast.ai, 01.11, 18:16 UTC [R]). The rate covers the GPU and the disk required by a given run. The disk fee is not negligible: at 250 GB and 0,53 $/GB/month it adds 0,19 $/h to 0,80 $/h for two cards.

machine	disk	offers	min [$/h]	median [$/h]
8× RTX 4090	500 GB	6	4,19	6,56
8× RTX 5090	500 GB	27	4,34	6,42
2× RTX 4090	250 GB	36	0,79	1,09
1× RTX 4090	100 GB	40	0,39	0,51

Filter: verified offers, `reliability` $\geq$ 0,98, on-demand. The market changes fast: at 17:58 UTC the cheapest 8×4090 cost 2,94 $/h, and 18 minutes later 4,19 $/h. We did not measure Slayer's rate on the RTX 5090, so before choosing the machine we measure 10 minutes on both types (about 1 $).

Stage cost [estimate]. The time is the number of new tokens divided by 86,5 thousand tok/s, and the cost is the time times the 8×4090 rate from the table (min–median). Slayer 1.06 saw 14 524 133 431 tokens (section 1).

stage	new tokens	time	GPU cost
1.07 (today)	9 bn	29 h	121–190 $
to 40 bn total	25,5 bn	82 h	343–537 $
to 100 bn total	85,5 bn	275 h	1 150–1 800 $

Beyond the GPU there is also: evaluation on a separate 1×4090 card (about 12 full-board points at 989 s measured on the A100, that is about 3,3 h, plus 10 loss-only points for avg10; we did not measure the board rate on the 4090, so we assume 4–12 h, about 2–6 $ on 9 bn [estimate]), 10 minutes of rate measurement before the run (about 0,5–1 $) and data transfer (download 2,67 $/TB, upload 4 $/TB [R]). Data building and scanning run on the CPUs of the machine that is paid for anyway (below), so they do not add rental.

Batch 1 (probe 1.07b): final settlement. Instance 58683982: 2×4090, 250 GB disk, rate 0,985 $/h, ran from 16:45:10 UTC 01.11 to 03:09:18 UTC 02.11, that is 10 h 24 min. The container has a limit of **46 cores** (`cpu.max` 4608000/100000) and **181 GiB RAM** (`memory.max`), although `nproc` and `free` show 192 cores and 377 GB of the host (read 19:21 UTC). In the working state the balance dropped at a rate of 0,997 $/h. During data download the rate was higher, about 1,2 $/h, due to transfer fees [S]. Balance: 12,411 $ at the start, 1,944 $ after deletion [K]. **Cost of the entire batch 1: 10,47 $**, on average about 1,01 $/h with transfer. It covers the training of four arms, the full evaluation of the arms and of the 1.06 point, the building and cleaning of the 1.07 data and the leakage scans of the heldouts. The instance was deleted by the account guardian at the 2 $ balance threshold [K]. Copies of the data, results and weights were on the station from 02:11 UTC, so an earlier manual deletion would have saved about 0,9 $.

Data logistics.

- **The station's link uploads slowly:** 0,3 MB/s on one stream and 1,6 MB/s on twelve [N]. 18 GB of 1.07 data through the station is several hours. The data is therefore created on the vast machines and passes **from node to node** (a one-time read-only SSH key for a single directory, then `sha256sum -c`, conditions in section 4). Downloading from the node to the station is also slow on one stream (about 0,5 MB/s). In parallel, in 24 streams of 128 MiB each through an SSH proxy, it reaches about 7–8 MB/s: the 9 files of 1.07 data (17 GB) and the weights of four arms (17,2 GB) each passed in about 35–40 min [K, N].
- **CPUs of the batch 1 node:** training uses 2 of the container's 46 cores. On the rest, at priority `nice 19`, we build the 1.07 data and scan it for leakage without additional rental. The container's memory limit (181 GiB) was exceeded at 19:13 UTC by two replays at once with the EN composition (OOM, training survived), which is why the data runs and scans go one after another, up to about 40 processes in total [N].
- **We rent the** 8× **machine only once the data is ready and clean**, so that the cards do not sit idle. Scan rate: about 1 h per 9 bn [estimate, N].

- **1.07 machine disk: 500 GB.** The evaluation checkpoints and avg10 are 22 files, about 195 GB together (launcher configuration [N]), plus the cleaned data about 18 GB and its concatenated copy about 18 GB. The evaluation card pulls the checkpoints from the 1.07 machine (about 195 GB at 4 $/TB upload and 2,67 $/TB download, about 1,3 $ together [estimate]).

Cost safeguards. Each rented machine has two independent fuses: (1) on the node, launched as the first command (`bezpiecznik.py`, deletion of the instance with the instance key after a set time); (2) on the account side on the station (`straznik_wynajmu.py`): deletion after a time limit or when the balance is below 2 $. Manual deletion of the machine happens only after the results are downloaded and on the explicit order of the account owner. In batch 1 the balance threshold fired earlier than the time limit: the guardian deleted the instance at 03:09:18 UTC at a balance of 1,944 $, instead of at 03:37 UTC [K]. The final packing of the results was therefore scheduled for 02:30 UTC at the latest; the evaluation finished at 01:42 UTC.

What these numbers do not say. The efficiency of 8 cards (0,88) and the RTX 5090 rate are assumptions to be measured. Prices change within minutes. The costs of the 40 and 100 bn stages assume the same 1B model and the same training code.

11 Limitations: what the result does not prove

- **Noise is a lower bound.** z_k is a single realization: two checkpoints of one run, 48 steps apart. It does not cover seed noise or data ordering, so the true noise may be larger, and "movement" at $|\Delta|$ near $2\,z_k$ may turn out to be noise.
- **1.07 is not a controlled experiment.** It changes the data, their ordering and the number of steps all at once, and there is no control arm. The verdict "better than 1.06" concerns the whole setup and does not attribute the gain to any single decision (e.g. the EN share or the PL replay).
- **Small additions are unidentifiable.** A component on the order of 0,3 % of the mixture (F/K data from the probe) produces no separate, measurable effect in 1.07.
- **Test leakage.** The 13-gram scan does not detect paraphrases or translations of test items. Tasks flagged with leakage are read only on positions without hits, which reduces the power of the comparison. EWoK is not in the scan index, so it is not an axis.
- **Transfer from the literature.** The works in section 6 have a different scale, a different language or a different LR schedule than Slayer. Their results are a hypothesis for us, not an expected gain.
- **Probe 1.07b has low power.** The arm is 105 million tokens and 100 steps. No movement in the probe does not rule out an effect at the 1.07 scale, and movement in the probe does not guarantee it in 1.07. From the 15 tasks with leakage in the probe base we draw no conclusions, because the leakage is asymmetric between the arms (F and K swap in part of the C documents, and C2 is a different sample of the same pool).
- **The 200-item panel.** A "chance-level" classification in section 2 means that the panel showed no skill. It does not mean that the model lacks it or that the data cannot teach it.

12 After pretraining: SFT, preferences and verifiable correctness

Place in the setup. The 1.07 stage being prepared is further pretraining on text; corpus growth alone is not RLHF. After selecting the base checkpoint one can tune the way of answering: adhering to the instruction, the format, the language and the use of the skills already held. We do not claim that the preference stage replaces missing base data, nor that it cannot change knowledge results. The impact on each of these axes must be measured separately. The path below is a **plan for a separate stage**, not an additional process running in today's training.

SFT: demonstrations of correct behavior. Supervised fine-tuning uses instruction–response pairs. Ouyang et al., arXiv:2203.02155, describe demonstrations for SFT, then rankings of responses from humans and learning from such feedback. For Slayer, the candidate is its own verified PL/EN demonstrations plus human data with clear provenance. We do not copy formats or items from the tests on which we will measure the effect. We evaluate the stages separately: base, model after SFT and model after preferences. We do not attribute an improvement due to SFT to later RLHF.

RLHF: learning from human preferences. This stage requires different responses to the same instruction and a judgment of which a human considers better. The classical scheme fits a reward model to the rankings, and then updates the language model via reinforcement learning, with control of divergence from the reference model. This is not selecting the correct letter from a ready key.

Rafailov et al., arXiv:2305.18290, propose *Direct Preference Optimization* (DPO): direct optimization on preference pairs, without separately fitting a reward model and without sampling from the model during this optimization. This is a simpler candidate for a later attempt at preference learning, **not a synonym for the classical RLHF algorithm**. The authors' results on sentiment, summarization and dialogue are not proof of DPO's effectiveness on Slayer PL+EN, nor a cost estimate for our stage. We do not choose an algorithm solely by its name or a claim of simplicity.

Preference data: candidates, not a ready input set. The preliminary license qualification from 1 November points to human conversations and rankings from OpenAssistant OASST1/OASST2 (Apache-2.0) and human Aya data (Apache-2.0) for SFT. HH-RLHF (MIT) has human judgments and responses from Anthropic's model, with a provenance annotation in the card. This is a qualification of sources, not a receipt of a built dataset and not proof of sufficient PL coverage. The revision, generator provenance, subset, license and the R1 rule must be settled before use; the PII and decontamination rules from section 4 also apply.

The author's own rankings are useful especially for Polish. One can judge pairwise the responses of one's own Slayer, with the variant's identity hidden and the response order randomized before judging. Criteria for correctness, instruction fulfillment and style should be recorded separately, so that a preference for fluent style is not taken as an improvement in facts. The training and evaluation pools must be separated by document and task.

RLVR: verifiable correctness and format kept separate. DeepSeek-R1, arXiv:2501.12948v1, in §2.2 separates the rule-based correctness reward from the format reward. The checking rule can be deterministic, even though responses are sampled. GRPO is an update algorithm, and RLVR describes a kind of reward; they are not synonyms. In §2.4 the authors indicate that small distilled models were trained via SFT without RL. Their results **do not prove the effectiveness of direct RLVR** on an arbitrary small base.

For Slayer, RLVR remains a candidate for a separate attempt on tasks for which we have a verifier robust to workarounds, e.g. a computation or a code test. The verifier must distinguish correct content from merely correct notation. We do not run RLVR as a way to fix an unknown cause of a weak result, nor without independent evaluation tasks.

Condition for moving to the next stage. Before an attempt one must specify its target skill, the data and its provenance, the starting checkpoint, a control without the new stage, the budget and the evaluation rule. The measurement covers instruction following, verifiable correctness, blind preferences on held-out instructions, and the cost for PL/EN and base skills. The checkpoint is selected on validation, not by searching for the best test after the results. Preferences, format and knowledge have separate results; improving one of them is not automatically an improvement in the others. None of these proposals halts the construction of the 1.07 base data, nor does it mean that post-training has already been done.

13 Synthesis: package for the training team

State of execution. Data encoded with Slayer's tokenizer, a data-control procedure and a training and measurement design have been prepared. The 1.07 model training has not yet been performed. This document is material for review by the training team, not a report on model improvement nor approval to start a run.

What is prepared.

- **Post-control set v2:** 8 891 005 707 tokens in 9 files. The Polish web contains 3 110 430 927 tokens, the Polish replay 1 894 295 941, mixed English 1 650 618 814, and the additional FineWeb-Edu 2 235 660 025. The numbers describe the finished files, not the declared size of the sources.
- **Quality control:** personal-data masking, a scan of documents against the test indexes, removal of the indicated documents and a byte-for-byte comparison of the retained documents. The Polish web, after additional masking, passed a full re-scan. Token-to-source consistency samples were 1 000/1 000 for each checked file.
- **Addition after review:** the extra index covers 5 652 ARC-Easy EN, ARC-Easy-PL and English-original Belebele items. The total index coverage is 401 450 items. Additionally, 103 EN documents (421 063 tokens) were removed; the set total above reflects this change.
- **The replay is deliberate.** Exact copies of English documents from the 1.06 pool were excluded, but full deduplication of near copies was not performed. The disjointness of Polish-web identifiers is not proof of the absence of similar texts. This is how we bound the claim about data novelty.

Result of the completed probe. Two independent implementations agreed on the main reading of the F/K experiment at unchanged thresholds (section 7). F does not meet the format condition. K meets the rule of BPB movement of the correct answer downward, and the improvement also appears in the control group. This is a failure to confirm the predicted domain specificity, not a failure of the BPB metric. The finished set above remains unchanged; the decision to carry the data over into 1.07 belongs to the team. We do not turn the specificity prediction into a new success criterion after seeing the result.

What the review changed.

1. **A model result is not automatically a data effect.** A full run changes the mixture, the training length and the way the release is prepared. The comparison with arm B has the same budget only on the segment 13 848→13 998. The retained B results cover four axes: ARC-C cloze, MedMCQA cloze, MultiBLiMP and WikiText-2. For these axes one can compute the contrast Δ_{dane}; the other axes have no such control without an additional arm.

2. **The metrics were checked on a negative control.** The BPB of the correct answer changed strongly after further training without new data, so it is not a standalone verdict axis. Letter accuracy likewise remains descriptive. The evaluation family, the Holm correction and the noise model are defined by section 9.

3. **Selection was separated from evaluation.** The built H107 has two parts of 1 056 whole documents each: for checkpoint selection (1 042 856 tokens) and for the verdict (1 058 519 tokens). The split does not prove statistical independence. The built H' from outside HPLT has 1 561 documents and 1 050 043 tokens. The 9 files of 1.07 contain no full copies of it, but 256/1 561 documents (16,4 %) share a 13-gram; we do not declare full disjointness. The set `24c704ec` remains solely a descriptive retention measurement: parts of the core and legal data appear in the replay, and parts of the EN and web the 1.06 model saw earlier.

4. **We compare the same kind of artifact.** We compare the 1.07 model's avg10 with the 1.06 model's avg10, and a single checkpoint with a single one. An improvement due to averaging cannot be attributed to new data.

5. **LR decay is not an unexamined hypothesis.** After 150 steps the loss change relative to 1.06 was −0,0503911 nat for the arm with decay (A) and −0,0034372 nat for the constant LR (B). The A−B contrast is −0,0469539 nat; it is this that the rounded −0,0470 nat (SE 0,0006) corresponds to, not the change of A relative to 1.06. The gain for the full 1.07 stage and the comparison of its release branch with avg10 require a separate measurement.

Controls performed and remaining limitations. H107 is built and has matching manifest sums. The test against the 9 files of 1.07 v2 found no identical full documents nor shared 64-token prefixes among the 2 112 H107 documents. A descriptive scan showed a shared 13-gram in 681/2 112 H107 documents (32,24 %). This is the share of documents with a hit, not of shared tokens nor proof of memorization. The full H107 remains the main axis; for the same once-selected checkpoint we additionally report the result on documents with a hit and without one, without changing thresholds or re-selecting. We will check H' descriptively in the same way: its part without hits is 1 305 documents / 747 491 tokens with EOS; the full H' still remains an additional control that its subset does not replace. Copies of content from packs 0–7 of the 1.06 model were not checked. Remaining to be completed are the table of checkpoint noise and minimal detectable differences and the freezing of the final protocol. Merely writing the rule in the document does not mean these controls have been performed. Receipt of the v2 copy at the station was confirmed by an independent sum check 9/9 against the proper gate; the older v1 package is not its substitute. The HF archive remains a plan pending confirmation of the upload and the repository's privacy. We do not declare public redistribution nor availability on request.

Team decisions before the next run.

- approval of the mixture and budget and a possible release branch with decay;
- a decision on other format and knowledge inserts, taking the completed probe into account;
- a possible additional arm B' for the four axes without retained control results;
- qualification of further Polish sources (FinePDFs, ZPE, AGH textbooks), with measurements of quality, increment over 1.06, legal status, PII and leakage, instead of selection by volume alone;
- a separate plan for SFT and preferences after the base: they are not a completed stage of this preparation.